

CHEYNE HAY

**MULTIDISCIPLINARY BUSINESS SYSTEMS,
ENGINEERING & TECHNOLOGY PROFESSIONAL**

**Business Systems Engineer | Enterprise Solutions Architect | Software &
Automation Developer | Electrical, HVAC and Control Systems Specialist
| Technical Founder**

EXECUTIVE PROFILE

Cheyne is a multidisciplinary business systems, engineering and technology professional with experience spanning enterprise databases, software development, business operations, industrial automation, electrical control systems, HVAC and refrigeration engineering, applied physics, commercial strategy, cybersecurity, artificial intelligence and modern blockchain technologies.

His defining professional strength is the ability to understand and design **complete systems**.

Rather than treating software, engineering, automation, accounting, security and business operations as separate disciplines, Cheyne examines how they interact as parts of one commercial and operational ecosystem.

This systems-based approach allows him to move between business strategy, financial modelling, database architecture, physical machinery, electrical controls, software platforms, infrastructure and customer-facing products while retaining a clear understanding of the complete objective.

Cheyne's work is particularly suited to organisations, startups and technology projects that require someone capable of connecting:

- Commercial goals
- Business processes
- Financial controls
- Software systems
- Engineering infrastructure
- Automation
- Security
- Data
- Customer experience
- Long-term product development

His career has involved continual learning across several generations of technology, from enterprise database systems and programmable industrial controllers through to modern cloud applications, containerised infrastructure, artificial intelligence and decentralised Web3 platforms.

PROFESSIONAL IDENTITY

Cheyne is best described as a:

Multidisciplinary Business Systems Engineer

His specialisation is the design of practical systems that improve the way businesses operate.

This includes analysing how an organisation works, identifying inefficiencies or risks, designing technical and commercial improvements, and integrating the required software, hardware, controls, data and operational procedures.

His work often spans the complete lifecycle:

1. Understanding the commercial problem
2. Analysing the existing business process
3. Designing the operating model
4. Creating the financial and data structures
5. Architecting the technical platform
6. Developing or directing implementation
7. Integrating automation and infrastructure
8. Testing security and operational controls
9. Deploying the system
10. Monitoring performance and improving the platform

This end-to-end capability distinguishes Cheyne from professionals whose expertise is limited to one narrow technical area.

PROFESSIONAL DEVELOPMENT & TECHNOLOGY EVOLUTION

Cheyne's technical background began with formal Business Systems studies between 1997 and 1999.

This developed an early foundation in:

- Business analysis
- Information systems
- Organisational processes
- Database design
- Commercial data management
- Systems development
- Reporting
- Business administration
- Technology-supported operations

He subsequently studied and developed database systems used within civil and retail business environments.

This early work established a strong understanding of how organisational data, financial information, inventory, operations, customers and management reporting must interact within a practical commercial system.

During the early 2000s, Cheyne expanded into Java programming and Microsoft enterprise technology through MCSE-era systems learning.

This period contributed knowledge across:

- Java software concepts
- Enterprise computing
- Microsoft server environments
- Networking
- User and system administration
- Business software environments
- Client-server systems
- Technical troubleshooting
- Enterprise technology integration

Unlike someone entering software only through modern web development, Cheyne's understanding developed through the earlier foundations of enterprise computing, databases, networking, business systems and physical engineering.

This historical perspective helps him recognise which modern ideas are genuinely new and which are updated forms of proven systems concepts.

BUSINESS SYSTEMS ENGINEERING

Business systems engineering is the central discipline connecting Cheyne's experience.

He approaches a business as an interconnected system made up of:

- People
- Procedures
- Technology
- Finance
- Data
- Assets
- Customers
- Suppliers
- Security
- Governance
- Operational risks

A technical solution is only successful when it improves the performance of this broader system.

Cheyne's business systems capabilities include:

- Business requirements analysis
- Operational workflow mapping
- Process improvement
- Systems modelling
- Enterprise architecture
- Business data design
- Internal process automation
- Management reporting
- Digital transformation
- Technology selection
- Systems integration
- Operational risk identification
- Commercial platform design
- Long-term systems planning

He is able to translate business language into technical requirements and technical constraints back into clear commercial consequences.

This allows communication between business owners, accountants, technicians, engineers, software developers, infrastructure teams and customers.

BUSINESS DATABASE SYSTEMS & DATA ARCHITECTURE

Cheyne has studied and developed database systems supporting civil and retail business operations.

His understanding of database systems extends beyond simply creating tables or storing records.

It includes the relationship between:

- Business events
- Financial transactions
- Customers
- Inventory
- Jobs and projects
- Assets
- Suppliers
- Staff
- Access controls
- Reporting requirements
- Audit history

Core database and information-system capabilities include:

- Relational data modelling
- Business entity design
- Transactional workflows
- Data integrity
- Reporting structures
- Information classification
- Record traceability
- User permissions
- Operational dashboards
- Historical reporting
- Data migration planning
- Database-backed applications
- Business intelligence foundations

Cheyne understands that poorly designed data structures eventually create accounting problems, operational confusion, security weaknesses and unreliable reporting.

His systems designs therefore treat data architecture as a foundational business control rather than an afterthought.

SOFTWARE ENGINEERING & APPLICATION DEVELOPMENT

Cheyne has the ability to write and understand software code while also operating at the wider architectural and product level.

His software capabilities include:

- Application architecture
- Frontend and backend system design
- API design
- Business logic
- Database integration
- Authentication
- Permissions and role management
- Multi-user applications
- Subscription platforms
- Financial workflows
- Administration systems
- Reporting dashboards
- Modern web applications
- Containerised deployment
- Distributed services
- Integration with external systems
- Testing and iterative development

Cheyne generally approaches software from the perspective of the complete operating platform rather than an isolated feature or code file.

Questions central to his approach include:

- How does the system make money?
- How does data move through the platform?
- What happens when a service fails?
- How are users authenticated?
- How are administrators controlled?
- Can every transaction be traced?
- How will the system be upgraded?
- How will the platform scale?
- What happens when a provider changes?
- How is the user protected?
- How are business and financial risks reduced?

ENTERPRISE SOLUTION ARCHITECTURE

Cheyne's strongest software-related capability is enterprise and solution architecture.

He designs platforms by considering the complete relationship between:

- User interfaces
- APIs
- Databases
- Infrastructure
- Authentication
- Security
- Third-party providers
- Financial systems
- Administration
- Monitoring
- Deployment
- Recovery
- Commercial operations

His architecture work considers both immediate requirements and future development.

Core architectural principles include:

- Modularity
- Maintainability
- Clear system boundaries
- Secure access
- Data integrity
- Operational visibility
- Upgrade safety
- Failure recovery
- Provider independence
- Commercial scalability

He is particularly effective in environments where a system must integrate several complex domains instead of performing one narrow function.

HISTORICAL AND MODERN TECHNOLOGY CAPABILITY

Cheyne's technology knowledge spans several generations.

Earlier and foundational technologies

- Business information systems
- Relational database systems
- Java
- Microsoft enterprise systems
- MCSE-era infrastructure concepts
- Client-server architecture
- Business networking
- Enterprise user management
- Traditional software deployment
- PLC control environments
- Industrial sensors
- Building controls

Modern application technologies

- Web applications
- APIs
- Distributed services
- Linux
- Docker
- Containerised platforms
- Reverse proxies
- Cloud and self-hosted infrastructure
- Automated deployment
- SaaS systems
- Artificial intelligence-assisted engineering
- Modern authentication
- Real-time communication systems

Emerging technologies

- Blockchain
- Web3 applications
- Solana
- EVM ecosystems
- Wallet integration
- Digital assets
- Cross-chain systems
- Decentralised applications
- Creator platforms

- Tokenised services
- Subscription and payment ecosystems

This technology range allows Cheyne to assess modern systems using both current capabilities and long-established engineering principles.

ELECTRICAL ENGINEERING & CONTROL SYSTEMS

Electrical engineering is a major part of Cheyne's technical foundation.

Between approximately 2012 and 2018, he actively studied and completed electrical engineering trade-related learning and certification pathways.

His electrical and controls experience supports his work in:

- Electrical diagnostics
- Control circuits
- Industrial systems
- Instrumentation
- Sensors
- Automation hardware
- Embedded controllers
- Equipment integration
- Building control systems
- Electrical safety concepts
- Fault identification
- System commissioning
- Preventative maintenance

Cheyne understands both the digital decision-making layer and the physical electrical systems carrying out those decisions.

This is particularly valuable when developing automation, monitoring, environmental control, robotics and industrial systems.

HVAC, REFRIGERATION & BUILDING SYSTEMS ENGINEERING

Cheyne is a specialist trade-qualified HVAC engineer with practical understanding of heating, ventilation, air conditioning, refrigeration and environmental control systems.

His HVAC capability is not an incidental addition to his technology background.

It is one of the disciplines that developed his understanding of:

- System behaviour
- Energy transfer
- Control loops
- Pressure
- Temperature
- Fluid and gas movement
- Mechanical efficiency
- Fault diagnosis
- Preventative maintenance
- Environmental optimisation

Areas of capability include:

- Commercial HVAC systems
- Refrigeration systems
- Air-conditioning systems
- Building environmental control
- Mechanical services
- Equipment diagnostics
- Commissioning
- System optimisation
- Energy efficiency
- Preventative maintenance planning
- Sensor-based control
- Building automation integration

HVAC systems require electrical, mechanical, thermodynamic and control-system knowledge to work together.

This naturally reinforced Cheyne's multidisciplinary engineering approach.

INDUSTRIAL AUTOMATION, ROBOTICS & EMBEDDED SYSTEMS

Cheyne has worked with automation concepts from PLC-based systems of the 1990s through to modern embedded controllers and connected automation systems.

His automation knowledge includes:

- PLC logic
- Industrial controllers
- Control sequences
- Sensor integration
- Inputs and outputs
- Relays and actuators
- Robotics
- Process automation
- Building automation
- Environmental control
- Remote monitoring
- Embedded development
- Equipment coordination

Platforms and technology families include:

- PLC systems
- Siemens controllers
- Carel controllers
- ESP development boards
- Laboratory and prototyping boards
- Intelligent sensors
- Connected control devices
- Building management technologies

Cheyne can conceptualise and develop automated systems in which sensors collect information, logic determines the required response, and electrical or mechanical components perform the action.

He understands that effective automation is not simply about making machines move.

It involves:

- Reliable sensing
- Safe operating logic
- Fault detection
- Override capability
- Alarm handling
- Data collection
- Predictable recovery
- Human supervision
- Ethical and safe control

MATHEMATICS & ANALYTICAL REASONING

Mathematical reasoning is embedded throughout Cheyne's work.

His mathematical strengths support:

- Engineering calculations
- Financial modelling
- Systems analysis
- Control logic
- Optimisation
- Business forecasting
- Pattern analysis
- Algorithmic reasoning
- Data relationships
- Risk evaluation

Relevant areas include:

- Applied mathematics
- Algebraic reasoning
- Ratios and proportional relationships
- Engineering calculations
- Financial mathematics
- Statistical reasoning
- Systems modelling
- Logical analysis
- Pattern recognition
- Algorithmic structures
- Optimisation
- Numerical validation

Cheyne does not treat mathematics as an isolated academic subject.

He uses it as a practical tool for understanding relationships, predicting system behaviour, checking assumptions and identifying problems.

PHYSICS, THERMODYNAMICS & ENTROPY

Cheyne has a strong understanding and interest in physics, particularly where it relates to real engineering systems.

Areas of strength include:

- Thermodynamics
- Entropy
- Heat transfer
- Energy conversion
- Refrigeration cycles
- Pressure relationships
- Fluid behaviour
- Mechanical work
- Efficiency
- System equilibrium
- Control dynamics
- Energy losses

Thermodynamics influences his thinking beyond HVAC.

It provides a model for understanding how energy, information, cost and complexity move through systems.

Entropy is particularly relevant to his wider systems philosophy.

Without maintenance, discipline and corrective control, physical systems, software platforms, databases and businesses all tend toward increasing disorder.

Good engineering introduces the monitoring, controls and structure required to keep the system stable and useful.

ACCOUNTING & FINANCIAL SYSTEMS

Cheyne combines technical engineering capability with strong accounting and commercial reasoning.

His accounting-related capabilities include:

- Understanding income and expenditure
- Cash-flow awareness
- Cost analysis
- Budgeting
- Pricing models
- Revenue forecasting
- Subscription economics
- Profitability analysis
- Financial controls
- Transaction tracking
- Reconciliation concepts
- Business performance reporting
- Operational cost management

This enables him to design software and business systems that reflect genuine commercial requirements rather than purely technical functionality.

For example, when designing subscription platforms, marketplaces or payment systems, he considers:

- Gross revenue
- Fees
- Commission splits
- Customer balances
- Refund obligations
- Settlement
- Business liabilities
- Revenue recognition
- Treasury balances
- Audit trails
- Financial reconciliation

This financial perspective substantially improves the commercial quality of the systems he designs.

AUDITING, GOVERNANCE & BUSINESS ASSURANCE

Cheyne has a strong auditing mindset.

He naturally questions whether:

- A transaction can be traced
- A process can be verified
- An administrator's action is recorded
- A calculation can be reconciled
- A business rule is consistently enforced
- A failure can be diagnosed
- A reported number matches its source
- Permissions are properly controlled

His auditing and governance capabilities include:

- Process auditing
- Internal control assessment
- Transaction traceability
- Operational review
- Access review
- Data validation
- Reconciliation
- Risk identification
- Exception reporting
- Audit-log design
- Administrative accountability
- Business health review

This is not limited to financial auditing.

It applies equally to software, operations, automation, security and business processes.

BUSINESS PLANNING, STARTUPS & COMMERCIAL DEVELOPMENT

Cheyne is highly capable in business planning and startup development.

His approach combines product ideas with the financial, technical and operational systems required to make them sustainable.

Areas include:

- Startup concept development
- Business planning
- Market positioning
- Revenue-model design
- Product strategy
- Pricing
- Subscription plans
- Cost modelling
- Operational planning
- Growth strategy
- Risk analysis
- Customer journeys
- Technology commercialisation
- Business-health assessment

He understands that an idea is not yet a business.

A viable business requires:

- A defined customer
- A real problem
- A workable solution
- A sustainable revenue model
- Controlled costs
- Reliable operations
- Appropriate security
- Measurable performance
- A path to growth

Cheyne evaluates technology projects through this complete business lens.

BUSINESS HEALTH & OPERATIONAL IMPROVEMENT

Cheyne can assess the overall health of a business by examining how its systems interact.

Areas considered include:

- Revenue quality
- Cash flow
- Operating expenses
- Process efficiency
- Staff workload
- Customer retention
- Technology costs
- Operational risk
- Product quality
- Security weaknesses
- Data reliability
- Management visibility
- Scalability
- Technical debt

His goal is not simply to produce more reports.

It is to identify where a business is losing time, money, information, customer confidence or operational control and then design practical improvements.

ETHICAL SECURITY SYSTEMS

Security and ethics are central to Cheyne's system designs.

He does not treat security as a final feature added after development.

Security must exist throughout:

- Architecture
- Identity
- Permissions
- Data
- Infrastructure
- Financial transactions
- Administration
- Monitoring
- Recovery
- User protection

Core principles include:

- Secure-by-design architecture
- Least-privilege access
- Clear administrative controls
- Authentication
- Separation of roles
- Audit trails
- Financial traceability
- Data integrity
- Operational resilience
- Backup and recovery
- Responsible automation
- Transparent system behaviour

Ethical security also means designing systems that protect users from exploitation, hidden behaviour and unnecessary exposure.

Cheyne values:

- Consent
- Transparency
- Fair access
- Accurate reporting
- User control
- Responsible data handling
- Accountable administration
- Honest commercial design

He aims to build systems that are commercially effective without misleading or taking unfair advantage of their users.

INFRASTRUCTURE, DEPLOYMENT & DEVOPS

Cheyne has developed practical capability in modern infrastructure and deployment.

This includes:

- Linux environments
- Docker
- Containerised systems
- Multiple service architectures
- Reverse proxies
- Networking
- Domain routing
- Secure exposure of services
- Automated installation
- Upgrade scripts
- Monitoring
- Health checks
- Backup and restoration
- Recovery planning
- Production troubleshooting

He values deployment processes that are:

- Repeatable
- Documented
- Verifiable
- Recoverable
- Safe to upgrade
- Clear to operators

A software platform is not considered finished simply because the code runs locally.

It must also be deployable, maintainable, observable and recoverable in a real operational environment.

WEB3, BLOCKCHAIN & DIGITAL PLATFORMS

Cheyne's recent work includes the architecture and development of sophisticated Web3 and digital-platform concepts.

Areas include:

- Solana applications
- EVM ecosystems
- Multi-chain platforms
- Wallet connections
- Digital identity
- NFT marketplaces
- Token ecosystems
- Cross-chain transfers
- Swap and routing systems
- Creator platforms
- Subscription systems
- Digital payments
- Treasury models
- Custodial and non-custodial workflows
- Marketplace fees
- On-chain and off-chain integration

His approach applies traditional business, accounting, database and security principles to emerging technology environments.

Rather than treating blockchain as disconnected from ordinary business controls, he designs systems that still require:

- Reconciliation
- Permissions
- Accounting
- User support
- Security
- Monitoring
- Governance
- Commercial sustainability

COGNITIVE & ANALYTICAL STRENGTHS

Cheyne reports unusually strong recollection of:

- Events
- Dates
- Numbers
- System behaviour
- Technical details
- Recurring patterns
- Historical decisions
- Design changes

While he does not describe this as a perfect photographic memory, his recall and pattern-recognition ability support long-term systems work.

This enables him to identify when:

- A previous mistake is being repeated
- A requirement has been lost
- A system has regressed
- A number no longer reconciles
- A new design conflicts with an earlier decision
- A pattern indicates a deeper problem

These abilities are valuable in architecture, auditing, diagnostics, engineering and long-term product management.

PROFESSIONAL STRENGTHS

Cheyne's principal professional strengths include:

- Multidisciplinary systems thinking
- Business-first engineering
- Software and platform architecture
- Database and information systems
- Electrical and control-system reasoning
- HVAC and refrigeration engineering
- Industrial automation
- Robotics and embedded systems
- Applied mathematics
- Thermodynamics and physics
- Accounting and financial modelling
- Auditing and traceability
- Startup planning
- Business-health analysis
- Ethical security design
- Infrastructure and deployment
- AI-assisted engineering
- Web3 and blockchain platforms
- Pattern recognition
- Root-cause analysis
- Long-term product thinking

PROFESSIONAL PHILOSOPHY

Cheyne believes that technology should solve real problems and produce measurable value.

A well-designed system should be:

- Useful
- Honest
- Secure
- Maintainable
- Scalable
- Auditable
- Financially sustainable
- Understandable
- Recoverable
- Respectful of its users

Engineering is not merely the production of code, electrical circuits, machinery or documents.

It is the disciplined process of turning a complex problem into a reliable and understandable working system.

Cheyne's career has developed around this principle.

His value lies not only in knowing several disciplines, but in understanding how those disciplines connect.

He brings together business, finance, data, software, infrastructure, physical engineering, automation and security to design complete systems capable of operating in the real world.

PROFESSIONAL POSITIONING

Cheyne is appropriately positioned as a:

Multidisciplinary Business Systems Engineer and Enterprise Solutions Architect

Additional appropriate professional descriptions include:

- Technical Founder
- Systems Architect
- Business Technology Strategist
- Software and Automation Engineer
- Enterprise Systems Consultant
- Digital Platform Architect
- Industrial Controls and Automation Specialist
- Product and Commercial Systems Designer

His strongest professional value is in environments that require broad technical understanding, commercial awareness and the ability to connect multiple complex disciplines into one coherent solution.

CONCLUSION

Cheyne brings together nearly three decades of continuous learning and practical systems thinking across business, software, engineering and automation.

His professional background spans the evolution from 1990s business information systems and industrial controllers through enterprise databases, Java, Microsoft infrastructure, HVAC, electrical engineering and robotics to current artificial intelligence, cloud applications, Web3 and blockchain platforms.

This range is not a collection of unrelated skills.

Each discipline contributes to one central capability:

The ability to design complete, ethical and commercially practical systems.

Cheyne understands that successful technology must work at every level—from the sensor and electrical circuit through to the database, application, financial ledger, business process and end customer.

That integrated perspective defines his professional identity and provides the foundation for his continued work as a business systems engineer, technology architect, automation specialist and technical founder.